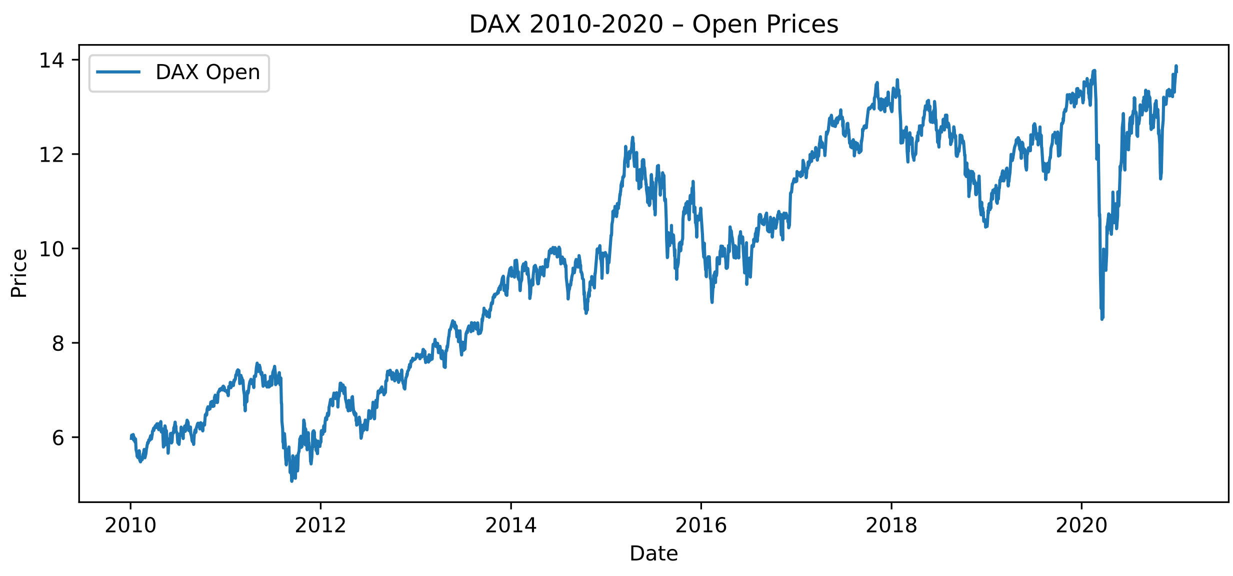
STATS 531 - Midterm Project

Steven Anthony Leone   Ethan Naegele

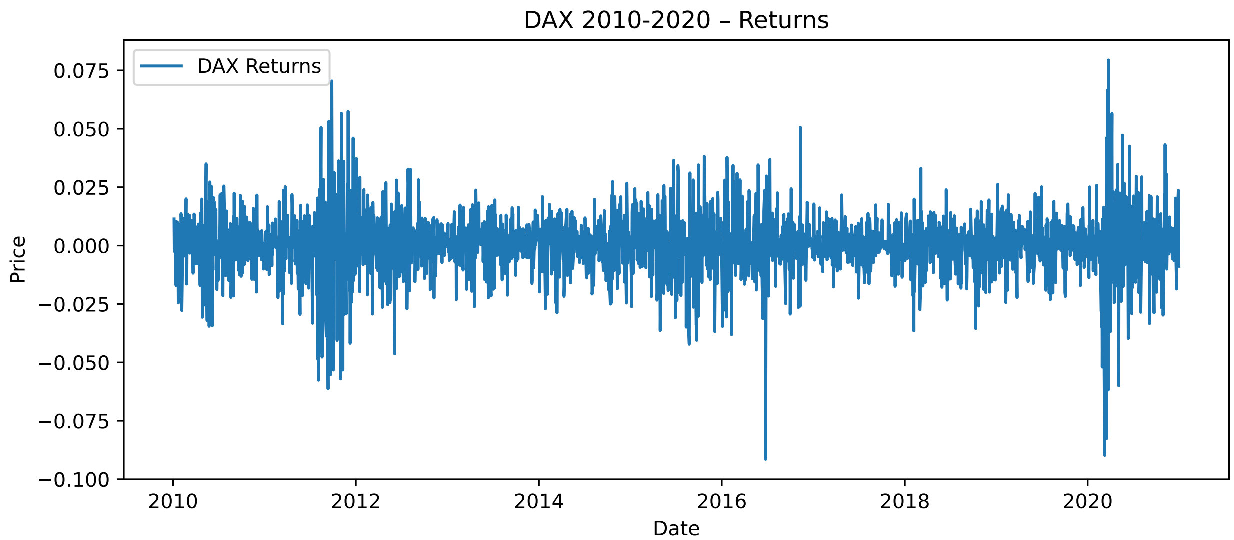
We look into the DAX Stock Index in Germany, and investigate how we can model the stock's opening prices, volatility, and other features. The DAX index tracks the performance of the 40 largest companies listed on the Regulated Market of the Frankfurt Stock Exchange (STOXX Ltd. 2026). They represent the diversified economy of Germany. It is generally known as the German equity benchmark, and has been around for over 30 years. We aim to model increases and decreases in the stock's open price and, as often seen in stock analysis, volatility.

Exploratory Data Analysis

We look at the daily stock price open for the DAX index for every day from 2010 to 2020 from a time series analysis perspective below.

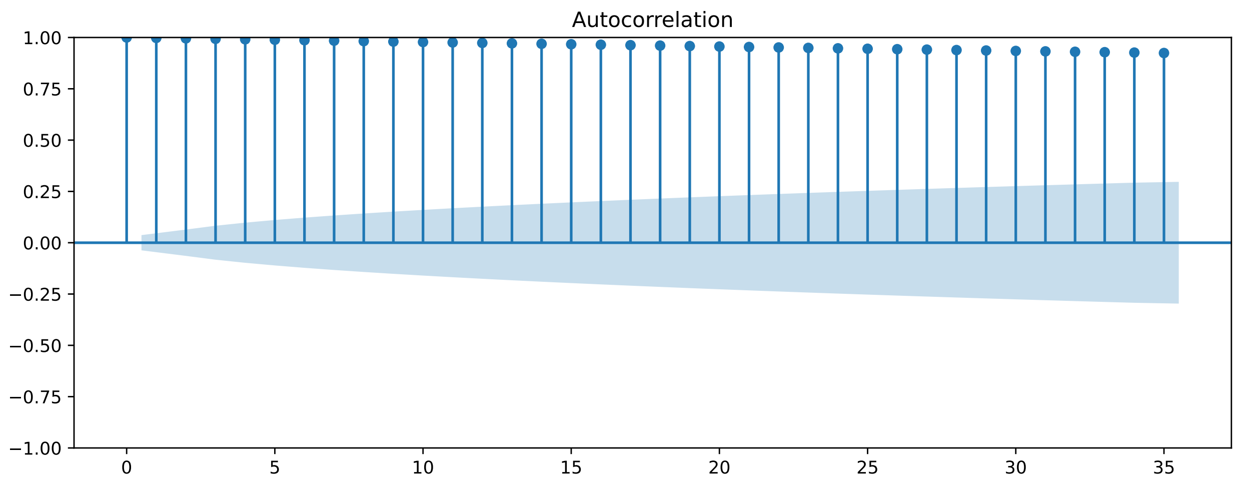


The data above for stock open shows a roughly nonstationary data. The mean is not constant, and we have a large dip in 2020, likely associated with covid. Below, we analyze our graph for volatility.

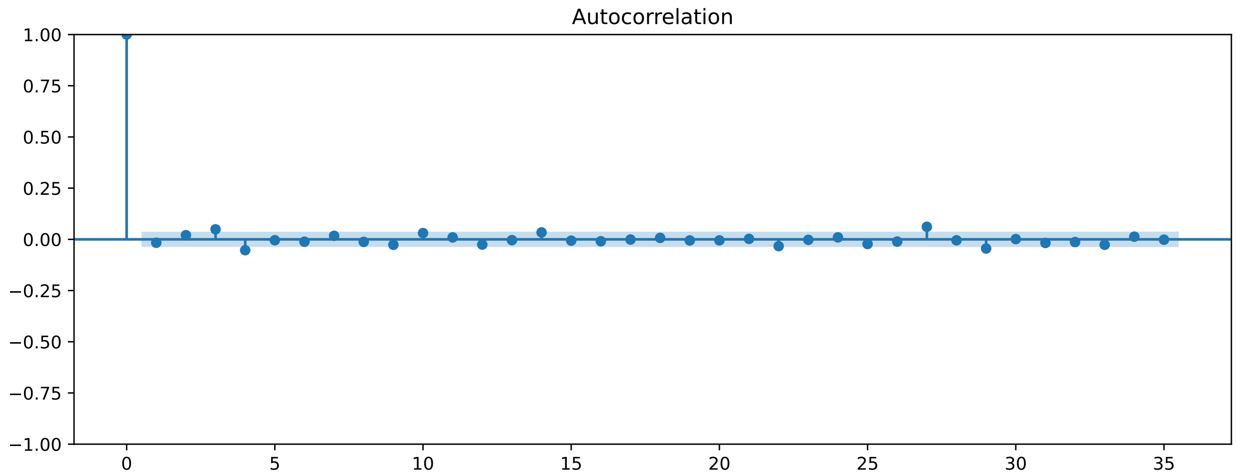


Our returns in the graph above follow a more stationary pattern, and follow a more or less constant mean.

We look at the autocorrelation function below to see how much lags depend on each other. We start with autocorrelation for DAX Open prices.

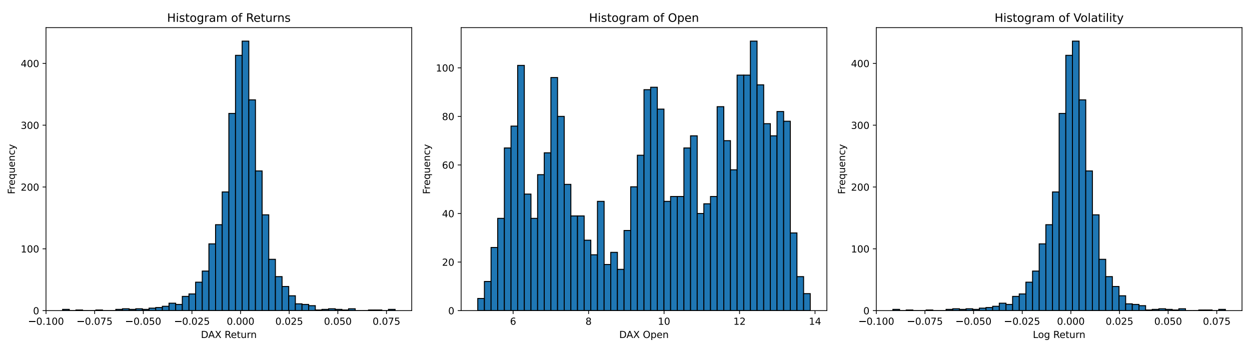


Opening prices are all correlated highly from the original value to down until the 35th lag. Correlation drops slowly. A model that depends on many of its lags seems suitable. The autocorrelation for DAX Returns is below.



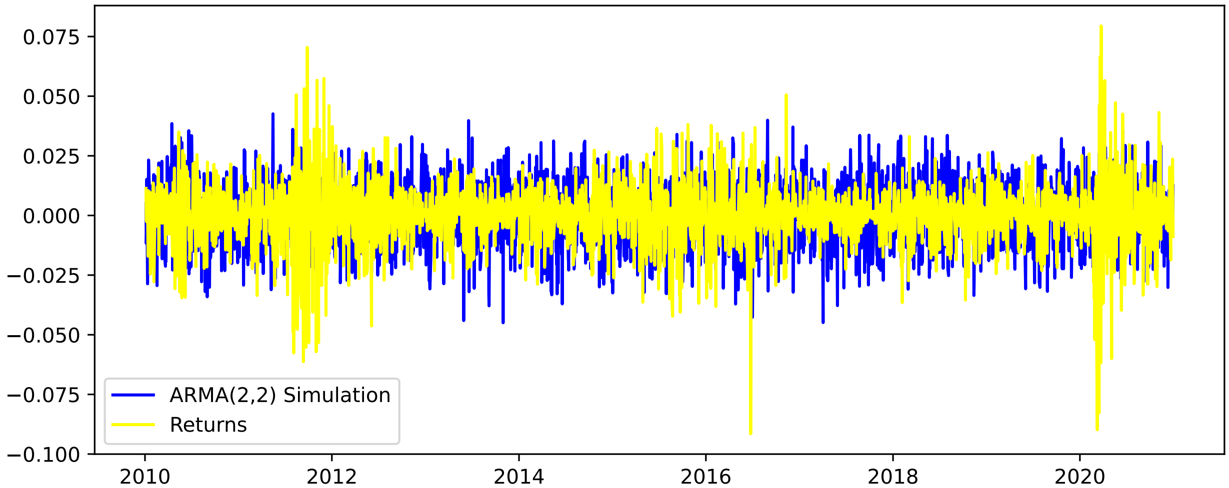
The returns are roughly i.i.d. They do not correlate much, and mostly fall within the shaded region. This means that the relationships between the first price return and its lags aren't as heavily correlated.

Now we show the histogram for the opening prices, returns, and volatility. These display the shapes of our data. While returns and volatility are normally distributed, opening prices are more uniformally distributed.



ARMA

Let's train a model to accurately reflect our observations from the exploratory data analysis. We start with an ARMA model.

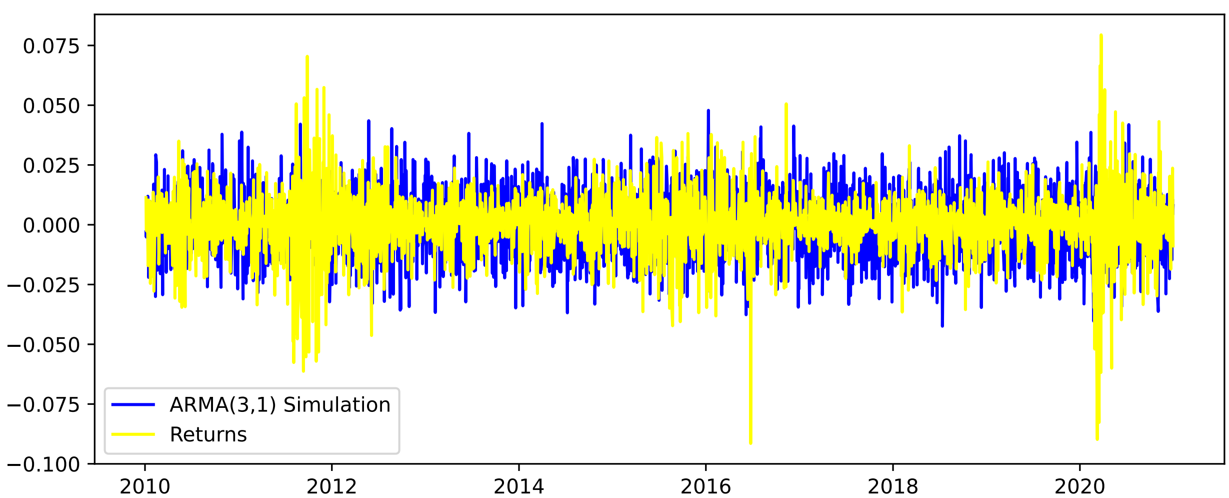


We look at the AIC values of the ARMA models below. The smallest AIC value is -3961.6, for ARMA (3, 1).

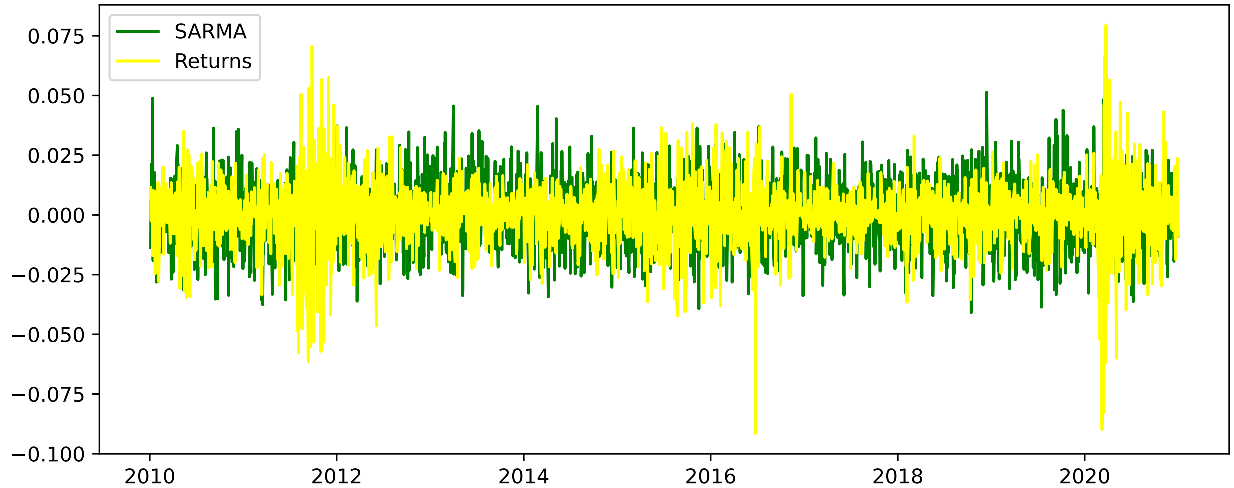
	MA(1)	MA(2)	MA(3)	MA(4)
AR(1)	-3965.5	-3965.7	-3972.9	-3973.5
AR(2)	-3962.0	-3966.6	-3972.8	-3971.7
AR(3)	-3961.6	-3970.7	-3974.7	-3972.2
AR(4)	-3972.8	-3974.8	-3972.4	-3968.0

Table 1: ARMA Model Comparison

We graph an ARMA(3, 1) model below. It is not much different from ARMA(2, 2).



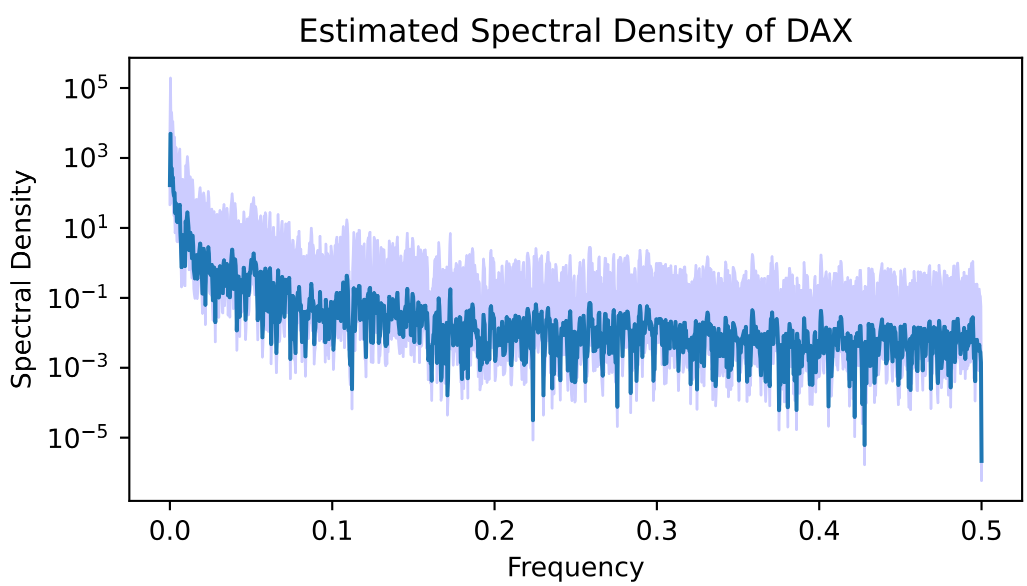
SARMA



Spectral Density Function

An estimator of the spectral density function determine a signal's frequency context content from a time series sample. In this section, we look at the frequencies for the DAX Stock Prices and Returns. It splits data into overlapping segments, computes modified periodograms, and averages them to reduce the variance. Advantages of Welch's method are that it provides a smooth and reliable estimate compared to alternatives.

Below is an estimate calculated through the Welch's method with a 95% confidence interval.



Appendix

TODO: Merge back in to SARMA

SARMA Model

The ARMA model can be written with a backshift operator, B, to represent the previous lag in a sequence. We can multiply a sequence by B in order to represent a lag farther back.

$$
\begin{aligned}
By_n &= y_{n-1} \\
B^2 y_n &= y_{n-2} \\
B^{12} y_n &= y_{n-12}
\end{aligned}
$$

When we have an ARMA model with white noise ϵ_t, we can represent the model simply with backshift operators and $\phi(B) = (1 - \phi_1 * B - \phi_2 * B^2 - ...\phi_p * B^p)$.

$$
\begin{gathered}
y_n = \phi_1 * y_{n-1} + \phi_2 * y_{n-2} + ...\phi_p * y_{n-p} + \epsilon_n \\
y_n - (\phi_1 * y_{n-1} + \phi_2 * y_{n-2} + ...\phi_p * y_{n-p}) = \epsilon_n \\
y_n - (\phi_1 * B * y_n + \phi_2 * B^2 * y_n + ...\phi_p * B^p * y_n) = \epsilon_n \\
(1 - \phi_1 * B - \phi_2 * B^2 - ...\phi_p * B^p) y_n = \epsilon_n \\
\phi(B) y_n = \epsilon_n
\end{gathered}
$$

In a seasonal model, we may only represent seasonal terms, ex, every 12. We can easily do this with the backshift operator with four functions, representing it as

$$\Phi(B^1 2) * \phi(B) y_t = \theta(B) * \theta(B^1 2) * \epsilon_t$$

This example will give us regular lags, as well as lags for every 12 (seasonal lags). As stock changes prices rapidly, this is relevant to a DAX index. We test it out below.

References

STOXX Ltd. 2026. "DAX® Index." https://www.stoxx.com/index/dax/.